



WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **RENEWCOILS ENGINEERING AND SUPPLY CO. LTD.**

WPS No. **RNC-WPS-012**

Date **12-Apr-13**

Revision No. **0**

Supporting PQR No.(s) **RNC-PQR-012**

Date **12-Apr-13**

Type(s) **Manual**

Welding Process(es) **Gas Tungsten Arc Welding + Shield Metal Arc welding**

(GTAW+SMAW)

JOINT (QW-402)

Joint Design **Single V**

Backing (Yes) ☐ (No) ☒ X

Backing Material (Type) **N/A**

☐ Metal ☐ Nonfusing Metal

☐ Nonmetallic ☐ Other

Notes

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. **8** Group No.

1 to P-No. **8**

Group No. **1**

Material Specification Type and Grade

SA 312 Gr.TP 321/TP347/SA358TP321

SA 312 Gr.TP 321/TP347/SA358TP321

Chem. Analysis and Mech. Prop.

SA403 Gr.WP321/WP347

TO SA403 Gr.WP321/WP347

Thickness Range :

Base Metal :

Groove

5 mm.-52.38 mm.

Fillet

ALL

Pipe Dia. Range :

Groove

ALL

Fillet

ALL

Other

FILLER METALS (QW-404)

Spec. No. (SFA)

GTAW

SMAW

5.9

5.4

AWS No. (Class)

ER19-10H

E347-15

F-No.

6

5

A-No.

8

8

Size of Filler Metals

Ø 1.6 & 2.4 mm

Ø 2.6 , 3.2 , 4.0 mm.

Deposited Weld Metals

5 mm.

21.19 mm.

Thickness Range :

Groove

10 mm.

42.38 mm.

Fillet

All

All

Electrode-Flux (Class)

-

Flux Trade Name

-

Consumable Insert

-

Other

Bohler CN18/11-IG

Bohler FOX E347H

POSITION (QW-405)

Position(s) of Groove

ALL

Welding Progression

Up

X

Down

Position(s) of Fillet

ALL

Temperature Range

1080°C ±10°C

Time Range

Holding Time 1 Hour

Notes

Heating Rate = 220°C Max. & Air Cooling

PREHEAT (QW-406)

Preheat Temp.

Min.

30°C

Interpass Temp.

Max.

170°C

Preheat Maintenance

N/A

Notes

N/A

GAS (QW-408)

Percent Composition

Shielding

Gas(es)

(Mixture)

Flow Rate

Trailing

Backing

Argon

99.99%

10-20 L/Min

Argon

100.00%

10-20 L/Min



WELDING PROCEDURE SPECIFICATION (WPS)

WPS No. WPS-012 Rev. 0

ELECTRICAL CHARACTERISTICS (QW-409)

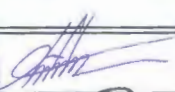
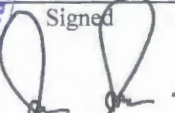
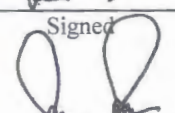
Current ☒ AC ☒ DC Polarity EN(GTAW) / EP(SMAW)
 Amps (Range) 80-120 Volts (Range) 10-15(GTAW) - 20-30(SMAW)
 Tungsten Electrode Size and Type Ø 2.4 mm, Thoriated (EWTh-2)
 (Pure Tungsten, 2% Thoriated, etc.)
 Mode of Metal Transfer for GMAW N/A
 (Spray arc, short circuiting arc, etc.)
 Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

String or Weave Bead BOTH APPLICATIONS
 Orifice or Gas Cup Size 8.0 mm - 19.0 mm
 Initial and Interpass Cleaning (Brushing, Grinding, etc.) BOTH APPLICATIONS
 Method of Back Gouging N/A
 Oscillation N/A
 Contact Tube to Work Distance N/A
 Multiple or Single Pass (per side) MULTIPASS
 Multiple or Single Electrodes SINGLE ELECTRODE
 Travel Speed (Range) 5-15Cm./Min.
 Peening N/A
 Other _____

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER19-10H	2.4 mm.	DCEN	80-120	10-15	5-15	
2st	GTAW	ER19-10H	2.4 mm.	DCEN	80-120	10-15	5-15	
3st & OVER	SMAW	E347-15	3.2-4.0 mm.	DCEP	60-120	10-15	5-15	

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of ASME-IX, 2010 latest Revision.

Prepared by: Mv. Chatchawan  30/04/13.
 Name Signed Date
 Reviewed by: ภูชิน สีหอนไชย  30/04/13
 Name Phuchin Seehomchai Signed Date
 Approved by: ภูชิน สีหอนไชย  30/04/13.
 Name Phuchin Seehomchai Signed Date

Procedure Qualification Record (PQR)

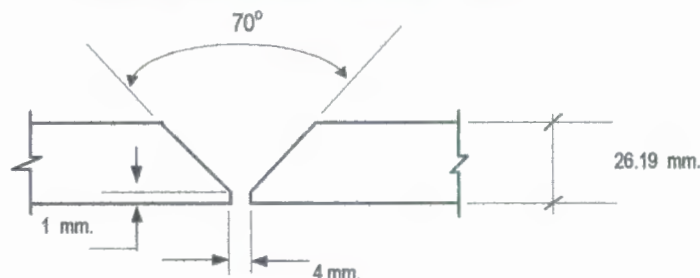
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : RNC-PQR-012

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-012	Date :	12 April 2013
WPS Number :	RNC-WPS-012		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW) + Shield Metal Arc Welding (SMAW)		
Type (Manual, Semi-auto, or Automatic) :	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)			POSTWELD HEAT TREATMENT (QW-407)		
Material Spec.	A358	To	A358	Temperature	1080° C
Type or Grade	TP321	To	TP321	Hold Time	1 Hour
P. No.	8	To	8	Others	Heating Rate 180 °C/Hr., Air Cooling
Thickness of Test Coupon		26.19 mm.			
Diameter of Test Coupon		Ø 16"			
Others					
FILLER METALS (QW-404)			GAS (QW-408)		
	GTAW	SMAW	Percent Composition		
SFA Specification	A 5.9	A 5.4		Gas (es)	Mixture
AWS Classification	ER 19-10H	E 347-15			Flow Rate
Filler Metal F-No.	6	5	Shielding	Argon	99.95%
Weld Metal Analysis A-No.	8	8	Trailing	-	-
Size of Filler Metal	Ø 2.4 mm.	Ø 3.2 mm.	Backing	Argon	99.95%
Others	Bohler CN18/11-1G	Bohler FOX E347H			25 L/Min.
Weld Metal Thickness	-		ELECTRICAL CHARACTERISTICS (QW-409)		
POSITION (QW-405)			Current	DC (GTAW)	DC (SMAW)
Position of Groove	6G		Polarity	EN	EP
Progression of Welding (Uphill or Down)	Up Hill		Amperage	113 - 134 (GTAW)	Voltage 10 - 12 (GTAW)
Others	N/A			101 - 115 (SMAW)	25 - 26 (SMAW)
PREHEAT (QW-406)			Tungsten Electrode Size	Ø 2.4 mm. 2% thoriated.EWTh-2	
Preheat Temperature	33° C		Others	N/A	
Interpass Temperature (max.)	166° C		TECHNIQUE (QW-410)		
Others	N/A		Travel Speed	4.65 - 10.17 cm/min.	
			String or Weave Bead	Both	
			Oscillation	N/A	
			Multiple or single Pass	Multiple Pass	
			Multiple or Single Electrodes	Multiple Electrodes	
			Others	N/A	

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number:- **RNC-PQR-012**

Tensile Test Report No.TS-13-04-093

Specimen	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (KN)	Ultimate Strength (N/mm ²)	Type of Failure & Loc.
1185/13 TR1	19.07	25.81	492.197	283.019	575.012	OUT OF WELD
1185/13 TR2	19.10	25.67	490.297	283.019	577.240	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-13-04-043

Type and Figure Number	Type of Bend	Indication	Results
1185/13 BS1	Transverse face bend	No Open Discontinuity	Accepted
1185/13 BS2	Transverse face bend	No Open Discontinuity	Accepted
1185/13 BS3	Transverse root bend	No Open Discontinuity	Accepted
1185/13 BS4	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170)

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Lateral Expansion		Drop Weight	
					% Shear	Mils	Break	No Break
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A
Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.PQR-006

Hardness Testing Report No. HV-13-04-024

Macrostructure Report No. MA-13-04-109

Impact Testing Report No.IM-13-04-032 (-20°C)

Welders Name Mr.Narongkorn Y./Mr.Charchai N. Welder Number -
Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor :

Date :



Inspected by : Mr.Wichit K.

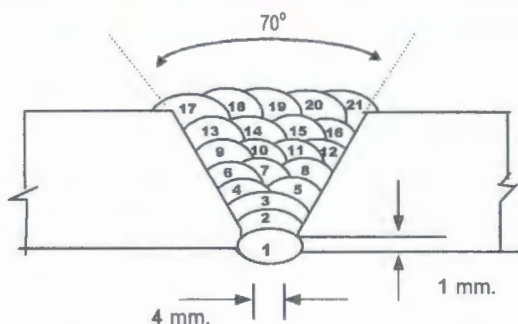
Certified by : Mr.Chaloemkiet R.

Date : 27 April 2013



	WELDING PROCEDURE	PQR No. : RNC-PQR-012
		WPS No. : RNC-WPS-012
	QUALIFICATION RECORDS	THIS PQR RECORDS ACTUAL TEST CONDITION AND RESULT

Welding Sequences for RNC-PQR-012 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or	Wire Feed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
							Weave Bead	Speed				
1	GTAW	ER 19-10H	2.4	113	10	DCEN	-	-	4.65	14.58	11	25
2	GTAW	ER 19-10H	2.4	134	12	DCEN	-	-	6.65	14.51	11	25
3	SMAW	E 347-15	3.2	101	25	DCEP	-	-	7.84	19.33	-	-
4	SMAW	E 347-15	3.2	105	25	DCEP	-	-	8.18	19.26	-	-
5	SMAW	E 347-15	3.2	115	26	DCEP	-	-	7.02	25.56	-	-
6	SMAW	E 347-15	3.2	110	26	DCEP	-	-	7.30	23.51	-	-
7	SMAW	E 347-15	3.2	110	26	DCEP	-	-	6.24	27.50	-	-
8	SMAW	E 347-15	3.2	115	26	DCEP	-	-	6.05	29.66	-	-
9	SMAW	E 347-15	3.2	111	26	DCEP	-	-	7.37	23.50	-	-
10	SMAW	E 347-15	3.2	114	26	DCEP	-	-	8.25	21.56	-	-
11	SMAW	E 347-15	3.2	115	26	DCEP	-	-	7.01	25.60	-	-
12	SMAW	E 347-15	3.2	112	26	DCEP	-	-	9.45	18.49	-	-
13	SMAW	E 347-15	3.2	115	26	DCEP	-	-	7.24	24.78	-	-
14	SMAW	E 347-15	3.2	115	26	DCEP	-	-	8.80	20.39	-	-
15	SMAW	E 347-15	3.2	115	26	DCEP	-	-	7.65	23.45	-	-
16	SMAW	E 347-15	3.2	115	26	DCEP	-	-	7.43	24.15	-	-
17	SMAW	E 347-15	3.2	112	26	DCEP	-	-	8.25	21.18	-	-
18	SMAW	E 347-15	3.2	112	26	DCEP	-	-	7.94	22.01	-	-
19	SMAW	E 347-15	3.2	112	26	DCEP	-	-	10.17	17.18	-	-
20	SMAW	E 347-15	3.2	112	26	DCEP	-	-	8.47	20.63	-	-
21	SMAW	E 347-15	3.2	112	26	DCEP	-	-	7.83	22.32	-	-



PAE TECHNICAL SERVICE CO.,LTD
NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250

Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 682208-9

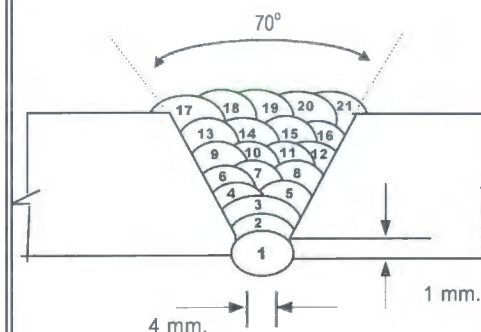
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 3

Client :	Project Name : MINI SHUTDOWN PTTGC	Location : RNC Work Shop	Report No. PQR-005
Renew Coils Engineering & Supply Co.,Ltd.	Customer Order No. : N/A	WPS No. : RNC-WPS-012	Date : 12-Apr-13

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
-	MR.NARONGKORN Y.	GTAW	6G	ASME IX	A358 TP321	ø 16" Thk. 26.19 mm.	A5.9	ACCEPTED	ACCEPTED
-	MR.CHERCHAI N.	SMAW					A5.4		

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	GTAW	ER 19-10H	2.4	DCEN	113	10	4.65	33	14.58	Argon	11	Argon	25
2	GTAW	ER 19-10H	2.4	DCEN	134	12	6.65	59	14.51	Argon	11	Argon	25
3	SMAW	E 347-15	3.2	DCEP	101	25	7.84	82	19.33	-	-	-	-
4	SMAW	E 347-15	3.2	DCEP	105	25	8.18	96	19.26	-	-	-	-
5	SMAW	E 347-15	3.2	DCEP	115	26	7.02	141	25.56	-	-	-	-
6	SMAW	E 347-15	3.2	DCEP	110	26	7.30	148	23.51	-	-	-	-
7	SMAW	E 347-15	3.2	DCEP	110	26	6.24	155	27.50	-	-	-	-
8	SMAW	E 347-15	3.2	DCEP	115	26	6.05	153	29.66	-	-	-	-
9	SMAW	E 347-15	3.2	DCEP	111	26	7.37	158	23.50				
10	SMAW	E 347-15	3.2	DCEP	114	26	8.25	160	21.56				

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME :

DATE :

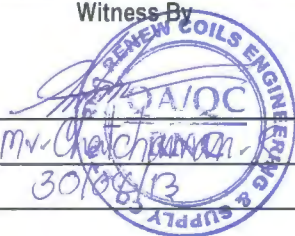


13-Apr-2013

SIGNATURE :

NAME :

DATE :



SIGNATURE :

NAME :

DATE :

Signature of Approved By

ภูชิน สีหอมไชย
Phuchin Seehomchai



PAE TECHNICAL SERVICE CO.,LTD

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250

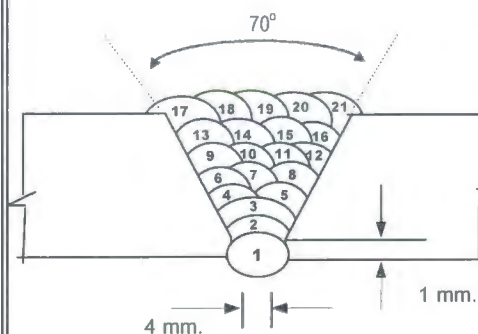
Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 682208-9

WELDER PROCEDURE QUALIFICATION RECORD

Page No. 2 of 3

Client : Renew Coils Engineering & Supply Co.,Ltd.		Project Name : MINI SHUTDOWN PTTGC			Location : RNC Work Shop		Report No. PQR-005		
		Customer Order No. : N/A			WPS No. : RNC-WPS-012		Date : 12-Apr-13		
Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
-	MR.NARONGKORN Y.	GTAW	6G	ASME IX	A358 TP321	ø 16" Thk. 26.19 mm.	A5.9	ACCEPTED	ACCEPTED
-	MR.CHERCHAI N.	SMAW					A5.4		

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
11	SMAW	E 347-15	3.2	DCEP	115	26	7.01	165	25.60	-	-	-	-
12	SMAW	E 347-15	3.2	DCEP	112	26	9.45	148	18.49	-	-	-	-
13	SMAW	E 347-15	3.2	DCEP	115	26	7.24	159	24.78	-	-	-	-
14	SMAW	E 347-15	3.2	DCEP	115	26	8.80	166	20.39	-	-	-	-
15	SMAW	E 347-15	3.2	DCEP	115	26	7.65	154	23.45	-	-	-	-
16	SMAW	E 347-15	3.2	DCEP	115	26	7.43	158	24.15	-	-	-	-
17	SMAW	E 347-15	3.2	DCEP	112	26	8.25	105	21.18	-	-	-	-
18	SMAW	E 347-15	3.2	DCEP	112	26	7.94	110	22.01	-	-	-	-
19	SMAW	E 347-15	3.2	DCEP	112	26	10.17	124	17.18	-	-	-	-
20	SMAW	E 347-15	3.2	DCEP	112	26	8.47	128	20.63	-	-	-	-

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME :

DATE :

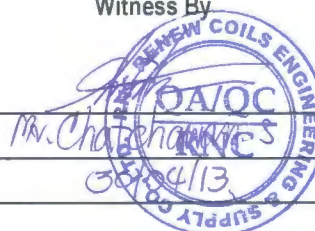


13-Apr-2013

SIGNATURE :

NAME :

DATE :



SIGNATURE :

NAME :

DATE :

(Signature)

ภูชิน สีหอมไชย

Phuchin Seehomchai

APPROVED BY

ภูชิน สีหอมไชย
Phuchin Seehomcha

Phuchin Seehomcha

POSITIVE MATERIAL IDENTIFICATION REPORT

Client : Renew Coils Engineering & Supply Co.,Ltd.	Analyzer Model : NITON XL3t 800	Test Date : 27-Apr-13
Project : Mini Shut Down PTTGC	Analyzer Serial No.: 39210	Report No. PMI-001
Procedure No. : PAE-PMI-007 Rev.0	Location : RNC SHOP	Attached Report : -
WPS No. RNC-WPS-012		

[illegible]

Total : 3 Points

NOTE :

- The Composition material Analysis was carried out on Surface of material
- The result of inspection accordance with ASTM table1 specification.

<u>Mat'l Spec. A358 TP321</u>	<u>Chemical Range Base 1 :</u>	Cr. 17.00 - 19.00	Ni. 9.00 - 12.00
	<u>Chemical Range Weld :</u>	Cr. 17.00 - 19.00	Ni. 9.00 - 13.00
	<u>Chemical Range Base 2 :</u>	Cr. 17.00 - 19.00	Ni. 9.00 - 12.00

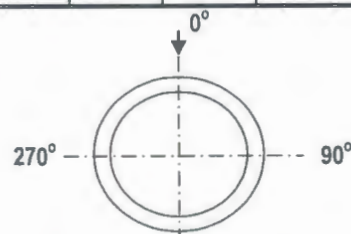
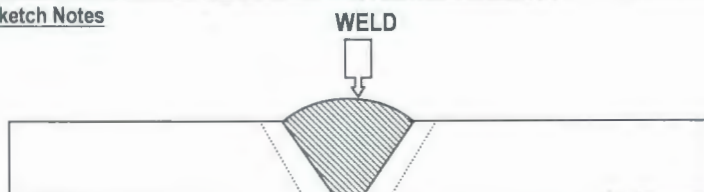
SIGNATURE :	PAE INSPECTED BY	REVIEWED / WITNESSED BY	APPROVAL BY
			
NAME :	Mr. Anuborn M. TECHNICAL SERVICE	Mr. Chaiyachon	ภูชน สีหอมไชย Puchin Seehomchai
DATE :	27-Apr-13	30/4/13	

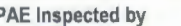
FERRITE CONTENT REPORT

Client :	Renew Coils Engineering & Supply Co.,Ltd.	Date :	27-Apr-13
Project :	Mini Shut Down PTTGC	Report No.:	FR-001
Project No.	PQR	Sheet :	1 OF 1
Acceptance Criteria :-	ASME B 31.3	Analyzer Model :	Feritscope SMP30
WPS No. :	RNC-WPS-012	Analyzer Serial No. :	021-26950A
Batch / Heat No. :	-	Material Specification / Grade :	A358 TP321

[illegible]

Weld Map/Sketch Notes



	PAE Inspected by		Approval by
Name	Mr. Anusorn M.		
Signature			ภูชิน สีหอมไชย
Date	27 Apr 13		Phuchin Seehomchai



Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnplm.co.th



0189

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-13-04-093

Revision No. 0

Test Product	PQR No. RNC-PQR-012	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A358 TP321	Type of Test	Reduced Section Tension
Dimension	Ø 16" #100 (26.19 mm Thk.)	Equipment's Capacity	200 TONS
Date Received	23 April 2013	Equipment / Serial No.	SHIMADZU UH200A / 32098926
Date Tested	25 April 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	1185/13 TR1	N/A	25.81	19.07	492.197	N/A	N/A	N/A	283.019	575.012	N/A	OUT OF WELD
2	1185/13 TR2	N/A	25.67	19.10	490.297	N/A	N/A	N/A	283.019	577.240	N/A	OUT OF WELD

The unit of N/mm² is equivalent to MPa

Additional details : WPS No. RNC-WPS-012

Welding Process : GTAW+SMAW

Filler/Electrode Spec't : ER19-10H E347-15

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 26 / 04 / 13



Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 26 / 04 / 13

LPN PLATE MILL

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Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnplm.co.th



Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.
Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Page 1 of 1
Report No. BD-13-04-043
Revision No. 0

Test Product	PQR No. RNC-PQR-012	Material Characterization	Pipe
Project Name	N/A	Test Method	ASME Sec. IX : 2010 edition
Project No.	N/A	Standart Of Test	ASME Sec. IX : 2010 edition
Material Specification	A358 TP321	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 16" #100 (26.19 mm Thk.)	Diameter of Meudrel	40 mm
Date Received	23 April 2013	Bend Angle	180 Degree.
Date Tested	25 April 2013	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	1185/13 BS1	Transverse side bend	9.90x24.69x161.42 mm	No Open Discontinuity	N/A	-
2	1185/13 BS2	Transverse side bend	9.96x24.67x162.25 mm	No Open Discontinuity	N/A	-
3	1185/13 BS3	Transverse side bend	10.12x24.59x162.90 mm	No Open Discontinuity	N/A	-
4	1185/13 BS4	Transverse side bend	10.13x24.59x161.04 mm	No Open Discontinuity	N/A	-

Additional details : WPS No. RNC-WPS-012

Welding Process : GTAW+SMAW

Filler/Electrode Spec't : ER19-10H E347-15

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 26/04/13

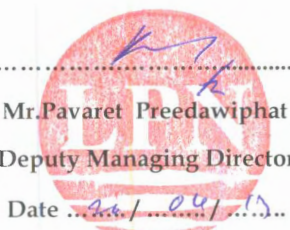


Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 26/04/13



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Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnplm.co.th



Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD
Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Page 1 of 1
Report No. IM-13-04-032
Revision No. 0

Test Product	PQR No. RNC-PQR-012	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 and ASTM E23M-07a ¹
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A358 TP321	Type of Test	Charpy V-Notch Impact
Dimension	Ø 16" #100 (26.19 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	23 April 2013	Equipment 's Capacity	542 Joules
Date Tested	25 April 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Specimen Dimension			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	1185/13-1	-20°C	10.009	10.004	54.85	273.99	296.20	Base Metal	-
2	1185/13-2	-20°C	10.008	10.000	54.87	310.77			
3	1185/13-3	-20°C	10.000	10.000	54.87	303.84			
4	1185/13-1	-20°C	10.003	10.008	54.88	224.00	241.68	HAZ	-
5	1185/13-2	-20°C	10.001	10.003	54.87	258.50			
6	1185/13-3	-20°C	10.003	10.006	54.89	242.54			
7	1185/13-1	-20°C	10.012	10.001	54.86	122.65	124.62	Weld Metal	-
8	1185/13-2	-20°C	10.004	10.000	54.89	130.61			
9	1185/13-3	-20°C	10.008	10.007	54.91	120.61			

Additional details : WPS No. RNC-WPS-012

Welding Process : GTAW+SMAW

Filler/Electrode Spec't : ER19-10H E347-15

Reported by :
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date 26 / 04 / 13



Approved by :
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date 26 / 04 / 13



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Macrostructure Analysis Report

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

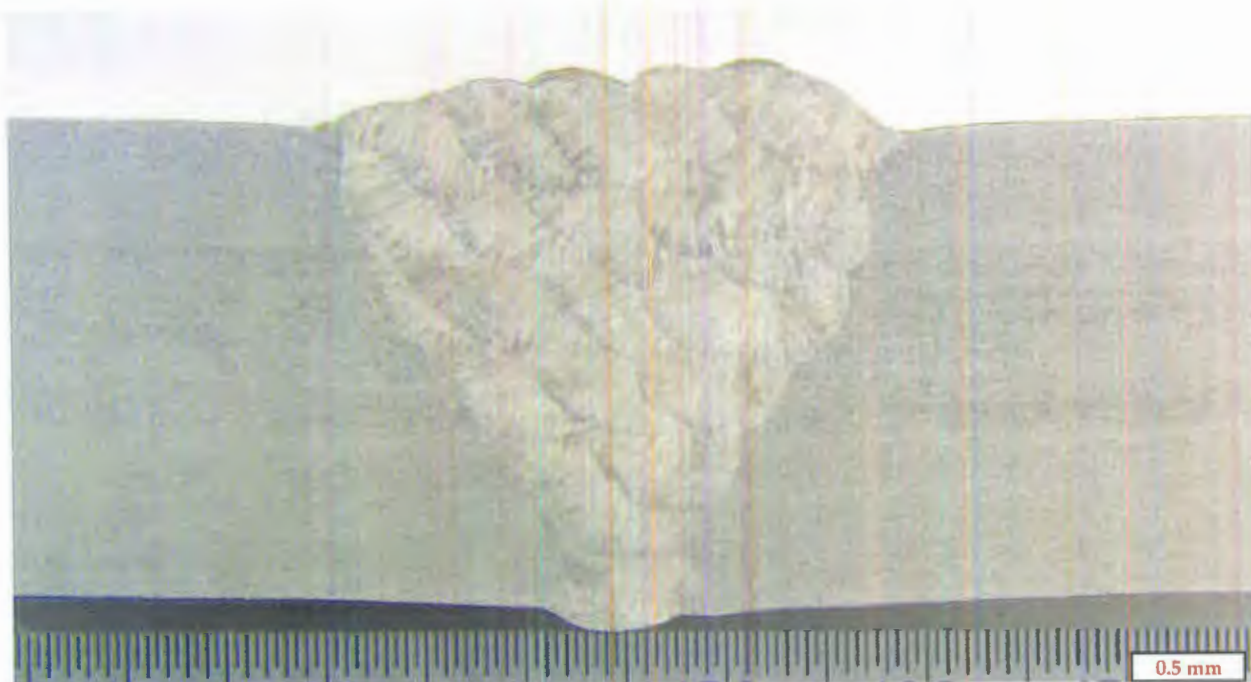
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Page 1 of 1

Report No. MA-13-04-109

Revision No.0

Test Product	PQR No.RNC-WPS-012	Material Characterization	Pipe to Pipe
Test No.	1185/13 MA	Equipment used	Microscope upto 40X (times)
Test Location	-	Serial No.	LEICA 0650854912
Material Specification	A358 TP321	Method of Preparation	ASME Section IX : 2010 edition
Dimension	Ø16"#100 THK26.19	Test Method	ASME Section IX : 2010 edition
Welding Process/Position	GTAW+SMAW	Standard of Test	ASME Section IX : 2010 edition
Type of Joint	Butt Joint	Etchant Reagent	25 ml HNO ₃ + 75 ml H ₂ O
Welder Name / No.	N/A	Time of Etching	5 minutes
Date Received	23 April 2013	Date Tested	25 April 2013



Macro-etch Examination : The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.

Additional details : -

Reported by :
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date : 26 / 4 / 13



Approved by :
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date : 27 / 4 / 13



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LPN PLATE MILL



Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

13307

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th

Hardness Testing Report

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Page 1 of 1

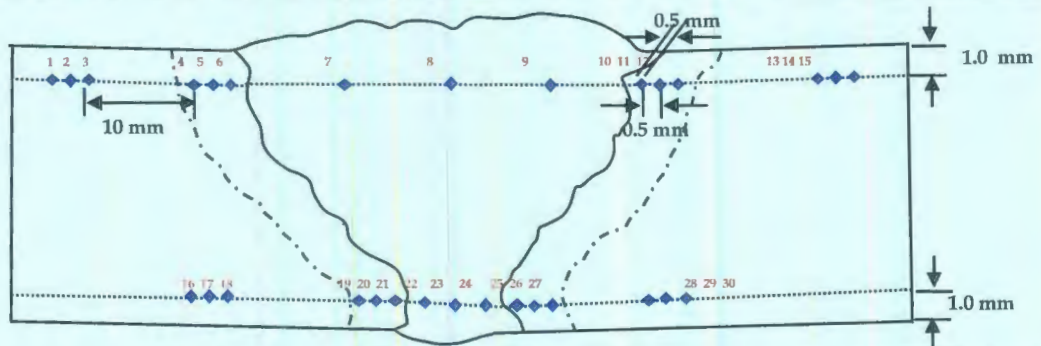
Report No. HV-13-04-024

Revision No. 0

Test Product	PQR No.RNC-WPS-017	Material Characterization	Pipe to Pipe
Test No.	I185/13 HN	Test Method	ASTM E92-82 (Reapproved 2003) ⁶²
Project Name	N/A	Standard of	ASME section IX: 2010 edition
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A358 TP321	Equipment	Wilson Wolpert , 432 SVD (V32D219)
Dimension	16"#100 THK26.19	Indenter Size	Pyramid diamond, angle of 136°
Welding Process/Position	GTAW+SMAW	Test Load	HV 10 Kgf.
Type of Joint	Butt Joint	Total Force Dwell Time	15 s
Date Received	23 April 2013	Cal. Block	No. A60023 No. A62072
Date Tested	25 April 2013	Ambient Temperature	26.9 °C

Test Location

Total = 30 points



Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale)
1	BASE	138.6	16	BASE	146.4	*****		
2	BASE	142.0	17	BASE	144.4			
3	BASE	139.6	18	BASE	145.4			
4	HAZ	135.0	19	HAZ	147.6			
5	HAZ	138.9	20	HAZ	147.7			
6	HAZ	135.5	21	HAZ	145.0			
7	WELD	174.8	22	WELD	170.1			
8	WELD	170.6	23	WELD	153.3			
9	WELD	178.2	24	WELD	173.5			
10	HAZ	135.8	25	HAZ	146.0			
11	HAZ	138.7	26	HAZ	147.2			
12	HAZ	138.1	27	HAZ	146.5			
13	BASE	132.8	28	BASE	139.3			
14	BASE	130.1	29	BASE	136.5			
15	BASE	133.6	30	BASE	137.4			

Additional details : -

Reported by :
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date : 26/04/13



Approved by :
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date : 26/04/13

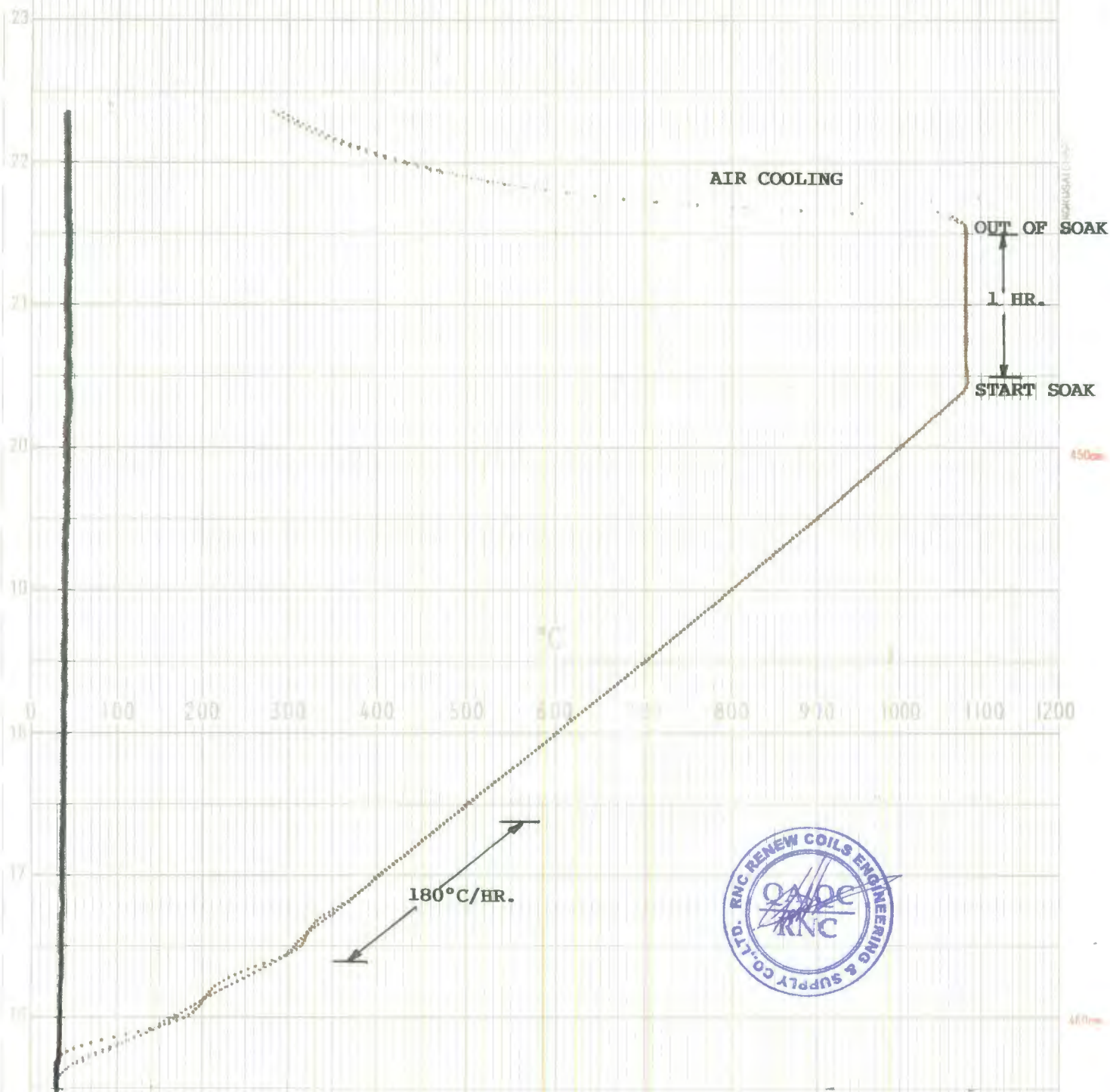
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LPN PLATE MILL

Completion Date: **13-04-2013** Procedure No.

	Global Heat Treatment (Thailand) Co., Ltd.			
	Checked By	Approved By	Approved by	Approved by
Signature:				
Name:	KOMKHUN D.	Chatchawan K.	ภูชิน สีหอมไชย	ภูชิน สีหอมไชย
Date:	13-04-2013	10/04/13	Phuchin Seehomchai	Phuchin Seehomchai



DATE. 13-04-2013

CHART NO. PW - 001

RECORDER NO. EH 07ZA194

CHART SPEED. 25mm./HR.

CUSTOMER. RENEW COILS ENGINEERING & SUPPLY CO., LTD.

PROJECT. MINI SHUT DOWN PTTGC

PQR NO. RNC-PQR-012

TC	COLOUR
1.	RED
3.	SKY BLUE
5.	BROWN

DATE. 13-04-2013

IC COLOUR

CHART NO. PW - 001

1. RED
3. SKY BLUE
5. BROWN

RECORDER NO. EH 07ZA194

CHART SPEED. 25mm./HR.

CUSTOMER. RENEW COILS ENGINEERING & SUPPLY CO., LTD.

PROJECT. MINI SHUT DOWN PTTGC

PQR NO. RNC-PQR-012

SIZE. 16" THK. 26.19 mm.

MATERIAL. SA312 GR.TP 321/TP347, SA403 GR.WP 321/WP347

START TIME. 11.00 AM.

TECHNICIAN. MR. WATCHARACHAI T.

